

ADVANCED COMPUTER NETWORKS

MODULE I

Overview of Computer Networks – Types, Topologies; Networking devices; OSI model, TCP/IP, Concept of Quality of Service (QoS), Client-server & Peer-to-peer architecture Application layer protocols- HTTP, FTP, POP3, and DNS

COMPUTER NETWORK

A Computer Network is a collection of interconnected devices that can communicate with each other and share resources such as data, applications, and hardware like printers, storage devices, etc.

The devices connected by network are called as **nodes**.

A node can be a device which is capable of sending or receiving data generated by other nodes on the network like a computer, printer etc. The links connecting the devices are called **Communication channels**.

Computer network is a telecommunication channel using which we can share data with other computers or devices, connected to the same network. It is also called **Data Network**. The best example of computer network is **Internet**.

A network must be able to meet certain criteria's, these are mentioned below:

- Performance
- Reliability
- Security

Performance of Computer Network

It can be measured in the following ways:

- **Transit time:** It is the time taken by a message to travel from one device to another.
- **Response time:** It is defined as the time elapsed between enquiry and response. Or, the total time, starting from sending the request up till receiving the response.

Other ways to measure performance are:

- Efficiency of software.
- Number of users using the network.
- Capability of connected hardware's.

Reliability of Computer Network

- It decides the frequency at which network failure occur. More the failures, less is the network's reliability.
- In order to improve the reliability of a computer network the infrastructure, devices, software, everything should be considered.

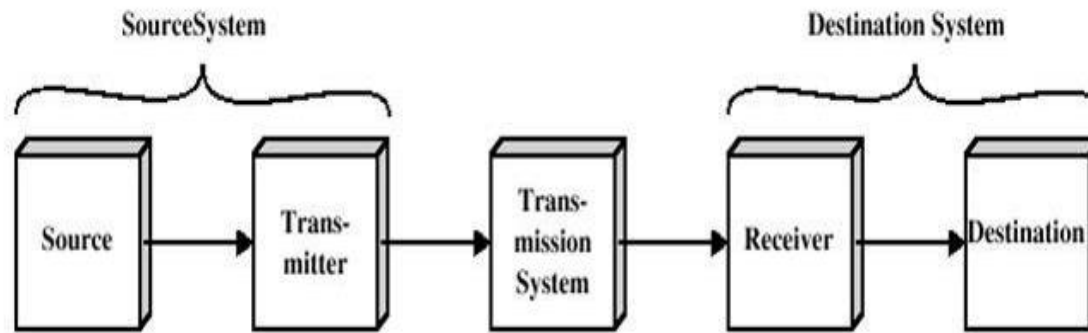
Security of Computer Network

- It refers to the protection of data from any unauthorised user or access. While travelling through the network, data passes many layers of network, during which data can be traced, if attempted. Hence security is also a very important characteristic for Networks.

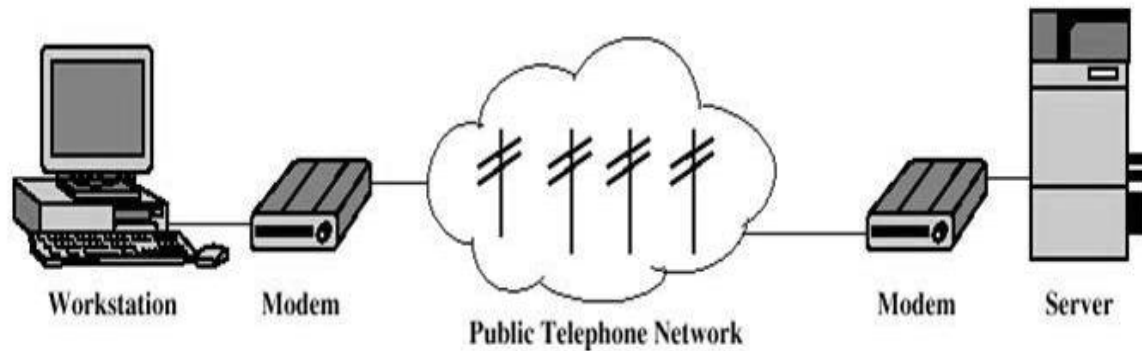
Properties of a Good Network

- **Interpersonal Communication:** We can communicate with each other efficiently and easily. Example: emails, chat rooms, video conferencing etc, all of these are possible because of computer networks.
- **Resources can be shared:** We can share physical resources by making them available on a network such as printers, scanners etc.
- **Sharing files, data:** Authorised users are allowed to share the files on the network.

Basic Communication Model



(a) General block diagram



(b) Example

- A Communication model is used to exchange data between two parties. For example: communication between a computer, server and telephone (through modem).
- **Source:** Data to be transmitted is generated by this device, example: telephones, personal computers etc.
- **Transmitter:** The data generated by the source system is not directly transmitted in the form it's generated. The transmitter transforms and encodes the data in such a form to produce electromagnetic waves or signals.
- **Transmission System:** A transmission system can be a single transmission line or a complex network connecting source and destination.
- **Receiver:** Receiver accepts the signal from the transmission system and converts it into a form which is easily managed by the destination device.
- **Destination:** Destination receives the incoming data from the receiver.

Data Communication

The exchange of data between two devices through a transmission medium is called **Data Communication**. The data is exchanged in the form of **0's** and **1's**. The transmission medium used is wire cable. For data communication to occur, the communication device must be a part of a communication system. Data Communication has two types - **Local** and **Remote** which are discussed below:

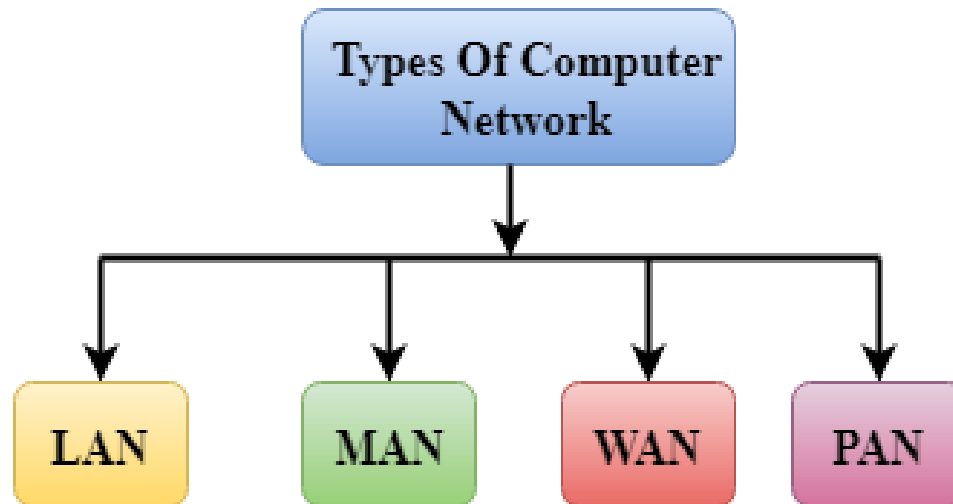
- **Local:** Local communication takes place when the communicating devices are in the same geographical area, same building, or face-to-face etc.
- **Remote:** Remote communication takes place over a distance i.e. the devices are farther. The effectiveness of a data communication can be measured through the following features:
 - **Delivery:** Delivery should be done to the correct destination.
 - **Timeliness:** Delivery should be on time.
 - **Accuracy:** Data delivered should be accurate

Components of Data Communication

- **Message:** It is the information to be delivered.
- **Sender:** Sender is the person who is sending the message.
- **Receiver:** Receiver is the person to whom the message is being sent to.
- **Medium:** It is the medium through which the message is sent. For example: A Modem.
- **Protocol:** These are some set of rules which govern data communication.

COMPUTER NETWORK TYPES

A computer network is a group of computers linked to each other that enables the computer to communicate with another computer and share their resources, data, and applications.



A computer network can be categorized by their size. A **computer network** is mainly of **four types**:

- LAN(Local Area Network)
- PAN(Personal Area Network)
- MAN(Metropolitan Area Network)
- WAN(Wide Area Network)

LAN (Local Area Network)



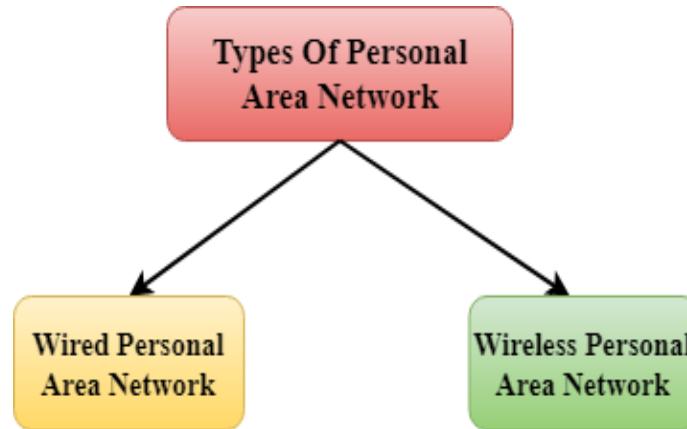
- Local Area Network is a group of computers connected to each other in a **small area** such as building, office.
- LAN is used for connecting **two or more personal computers** through a communication medium such as twisted pair, coaxial cable, etc.
- It is **less costly** as it is built with inexpensive hardware such as hubs, network adapters, and Ethernet cables.
- The data is transferred at an extremely **faster rate** in Local Area Network.
- Local Area Network provides higher security.

PAN (PERSONAL AREA NETWORK)



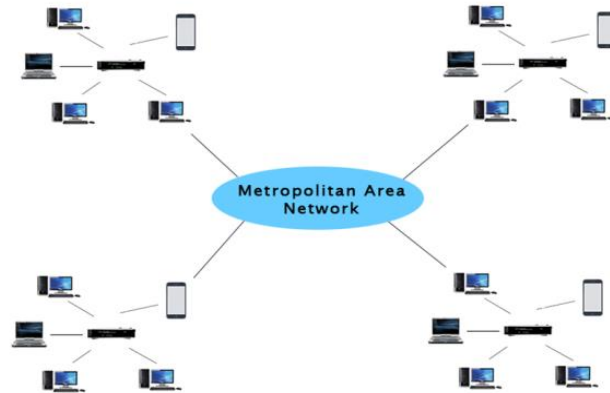
- Personal Area Network is a network arranged within an individual person, typically within a range of **10 meters**.
- Personal Area Network is used for connecting the computer devices of personal use is known as Personal Area Network.
- **Thomas Zimmerman** was the first research scientist to bring the idea of the Personal Area Network.
- Personal Area Network covers an area of **30 feet**.
- Personal computer devices that are used to develop the personal area network are the laptop, mobile phones, media player and play stations.

There are two types of Personal Area Network:



- **Wireless Personal Area Network:** Wireless Personal Area Network is developed by simply using wireless technologies such as WiFi, Bluetooth. It is a low range network.
- **Wired Personal Area Network:** Wired Personal Area Network is created by using the USB.

MAN(METROPOLITAN AREA NETWORK)

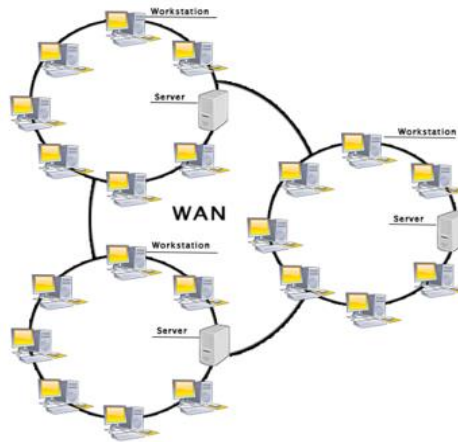


- A metropolitan area network is a network that covers a **larger geographic area** by interconnecting a different LAN to form a larger network.
- Government agencies use MAN to connect to the citizens and private industries.
- In MAN, various LANs are connected to each other through a telephone exchange line.
- It has a **higher range** than Local Area Network(LAN).

Uses of Metropolitan Area Network:

- MAN is used in communication between the banks in a city.
- It can be used in an Airline Reservation.
- It can be used in a college within a city.
- It can also be used for communication in the military.

WAN(WIDE AREA NETWORK)



- A Wide Area Network is a network that extends over a **large geographical area** such as states or countries.
- A Wide Area Network is quite **bigger network** than the LAN.
- A Wide Area Network is not limited to a single location, but it spans over a large geographical area through a telephone line, fibre optic cable or satellite links.
- The internet is one of the biggest WAN in the world.
- A Wide Area Network is widely used in the field of Business, government, and education.

NETWORK TOPOLOGY

Network Topology or Computer Network Topology is the description or blueprint of a network arrangement, connecting various nodes(sender and receiver) through lines of connection.



Point to Point



Bus



Ring



Star



Tree



Mesh



Hybrid

POINT TO POINT TOPOLOGY

Point-to-point topology is a type of topology that works on the functionality of the sender and receiver. It is the simplest communication between two nodes, in which one is the sender and the other one is the receiver. Point-to-Point provides high bandwidth.

Advantages

- **Security:** Point-to-point topology creates a private communication link between two nodes, making it difficult for other devices to access the network.
- **Fast data transmission:** Point-to-point topology eliminates delays and congestion by removing intermediate nodes.
- **Cost-effective:** Point-to-point topology requires minimal hardware, such as two network interface cards and a connecting cable.

Disadvantages

- **Limited scalability:** Point-to-point topology is not suitable for larger networks because it's difficult to connect more than a few nodes with a backbone cable.
- **High maintenance:** Point-to-point topology can be expensive to maintain and troubleshoot.
- **Limited fault tolerance:** Point-to-point topology is highly dependent on the common link, so if that link goes down, the entire network goes down.

BUS TOPOLOGY

Bus topology is a network type in which every computer and network device is connected to single cable. When it has exactly two endpoints, then it is called **Linear Bus topology**.

Features of Bus Topology

- It transmits data only in one direction.
- Every device is connected to a single cable

Advantages of Bus Topology

- It is cost effective.
- Cable required is least compared to other network topology.
- Used in small networks.
- It is easy to understand.
- Easy to expand joining two cables together.

Disadvantages of Bus Topology

- Cables fails then whole network fails.
- If network traffic is heavy or nodes are more the performance of the network decreases.
- Cable has a limited length.
- It is slower than the ring topology.

RING TOPOLOGY

It is called ring topology because it forms a ring as each computer is connected to another computer, with the last one connected to the first. Exactly two neighbours for each device.

Features of Ring Topology

- A number of repeaters are used for Ring topology with large number of nodes, because if someone wants to send some data to the last node in the ring topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node. Hence to prevent data loss repeaters are used in the network.
- The transmission is unidirectional, but it can be made bidirectional by having 2 connections between each Network Node, it is called **Dual Ring Topology**.
- In Dual Ring Topology, two ring networks are formed, and data flow is in opposite direction in them. Also, if one ring fails, the second ring can act as a backup, to keep the network up.
- Data is transferred in a sequential manner that is bit by bit.

Advantages of Ring Topology

- Transmitting network is not affected by high traffic or by adding more nodes, as only the nodes having tokens can transmit data.
- Cheap to install and expand

Disadvantages of Ring Topology

- Adding or deleting the computers disturbs the network activity.
- Failure of one computer disturbs the whole network.

STAR TOPOLOGY

In this type of topology all the computers are connected to a single hub through a cable. This hub is the central node and all other nodes are connected to the central node.

Features of Star Topology

- Every node has its own dedicated connection to the hub.
- Hub acts as a repeater for data flow.
- Can be used with twisted pair, Optical Fibre or coaxial cable.

Advantages of Star Topology

- Fast performance with few nodes and low network traffic.
- Hub can be upgraded easily.
- Easy to setup and modify.
- Only that node is affected which has failed, rest of the nodes can work smoothly.

Disadvantages of Star Topology

- Cost of installation is high.
- Expensive to use.
- If the hub fails then the whole network is stopped because all the nodes depend on the hub.
- Performance is based on the hub that is it depends on its capacity

MESH TOPOLOGY

It is a point-to-point connection to other nodes or devices. All the network nodes are connected to each other.

There are two techniques to transmit data over the Mesh topology, they are :

- Routing
- Flooding

Routing: In routing, the nodes have a routing logic, as per the network requirements. Like routing logic to direct the data to reach the destination using the shortest distance

Flooding: In flooding, the same data is transmitted to all the network nodes; hence no routing logic is required. The network is robust, and it's very unlikely to lose the data. But it leads to unwanted load over the network.

Types of Mesh Topology

- **Partial Mesh Topology :** In this topology some of the systems are connected in the same fashion as mesh topology but some devices are only connected to two or three devices.
- **Full Mesh Topology :** Each and every nodes or devices are connected to each other.

Features of Mesh Topology

- Fully connected.
- Robust.
- Not flexible.

Advantages of Mesh Topology

- Each connection can carry its own data load.
- It is robust.
- Fault is diagnosed easily.
- Provides security and privacy.

Disadvantages of Mesh Topology

- Installation and configuration is difficult.
- Cabling cost is more.
- Bulk wiring is required.

TREE TOPOLOGY

It has a root node and all other nodes are connected to it forming a hierarchy. It is also called hierarchical topology. It should at least have three levels to the hierarchy

Features of Tree Topology

- Ideal if workstations are located in groups.
- Used in Wide Area Network.

Advantages of Tree Topology

- Extension of bus and star topologies.
- Expansion of nodes is possible and easy.
- Easily managed and maintained.
- Error detection is easily done.

Disadvantages of Tree Topology

- Heavily cabled.
- Costly.
- If more nodes are added maintenance is difficult.
- Central hub fails, network fails.

HYBRID TOPOLOGY

It is two different types of topologies which is a mixture of two or more topologies. For example if in an office in one department ring topology is used and in another star topology is used, connecting these topologies will result in Hybrid Topology (ring topology and star topology).

Features of Hybrid Topology

- It is a combination of two or topologies
- Inherits the advantages and disadvantages of the topologies included

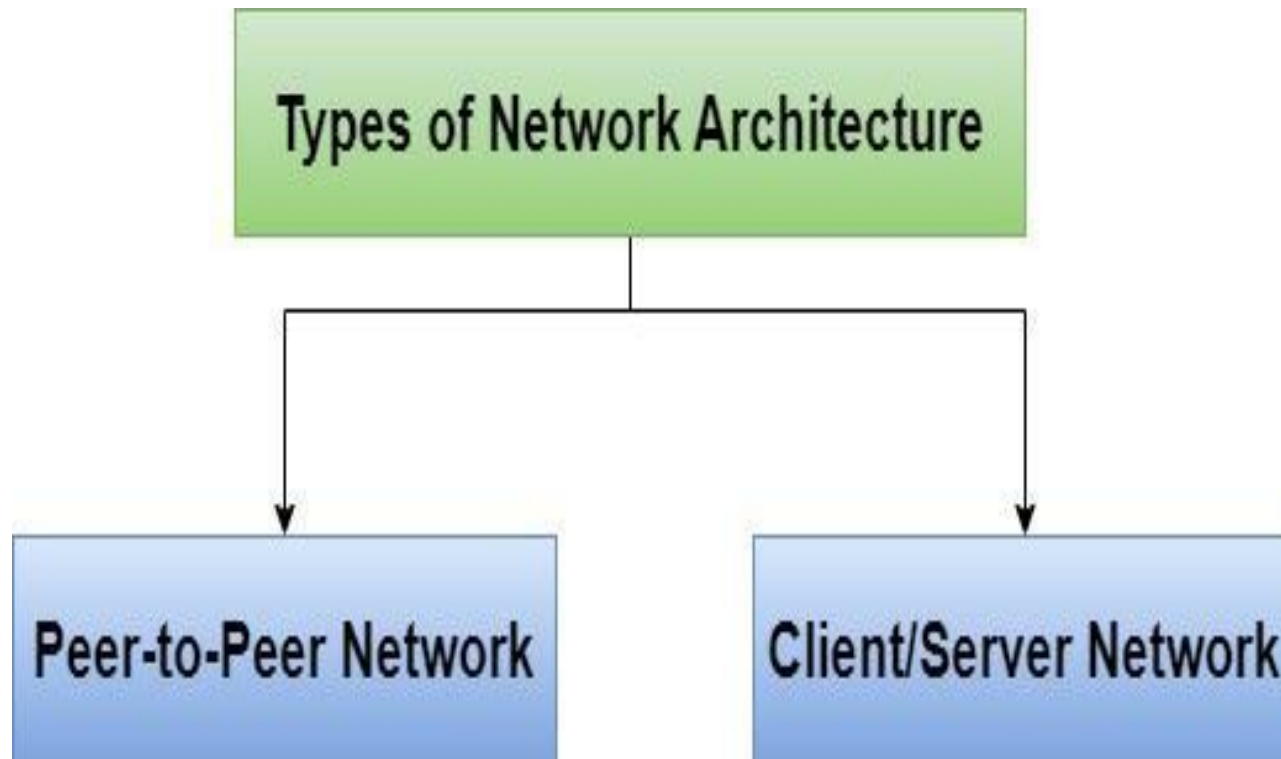
Advantages of Hybrid Topology

- Reliable as Error detecting and trouble shooting is easy.
- Effective.
- Scalable as size can be increased easily.
- Flexible.

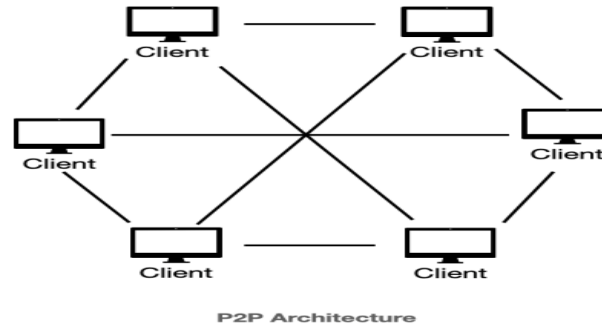
Disadvantages of Hybrid Topology

- Complex in design.
- Costly

NETWORK ARCHITECTURE



Peer-to-Peer Architecture



In a Peer-to-Peer (P2P) architecture, all nodes (peers) act as both clients and servers. Peers can request and provide services to each other without relying on a centralized server.

Characteristics

- Decentralized Network : No centralized server; all nodes are equal.
- Peer-to-Peer Communication : Peers communicate directly with each other.
- Equal Node Responsibility : All nodes are responsible for providing and requesting services.
- Self-Organizing : P2P networks can self-organize and adapt to changes.

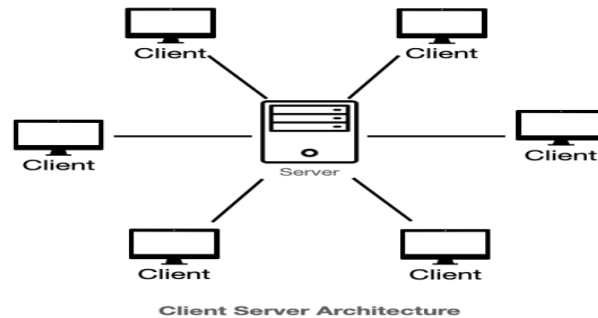
Advantages

- Decentralized : No single point of failure; network remains functional even if some nodes fail.
- Improved Scalability : P2P networks can handle a large number of nodes.
- Increased Flexibility : Peers can join or leave the network as needed.

Disadvantages

- Complexity : P2P networks can be more complex to manage and maintain.
- Security Risks : Decentralized nature of P2P networks can make them more vulnerable to security threats.

Client-Server Architecture



In Client-Server architecture, a centralized server provides services and resources to multiple clients. Clients request services or resources from the server, which processes the requests and sends responses back to the clients.

Characteristics

- **Centralized Server** : A single server manages and provides services to all clients.
- **Client-Server Communication** : Clients send requests to the server, which processes and responds to the requests.
- **Server-Client Relationship** : The server is responsible for managing and providing services to clients.
- **Scalability** : Client-Server architecture can be scaled up or down by adding or removing servers.

Advantages

- **Easy Maintenance** : Centralized server makes it easier to maintain and update services.
- **Security** : Server can implement security measures to protect data and services.
- **Scalability** : Can handle a large number of clients.

Disadvantages

- **Single Point of Failure** : If the server fails, all clients are affected.
- **Dependence on Server** : Clients rely on the server for services, which can lead to bottlenecks.

REFERENCE MODELS IN COMPUTER NETWORKS

The most important reference models are:

OSI reference model.

TCP/IP reference model.

OSI MODEL

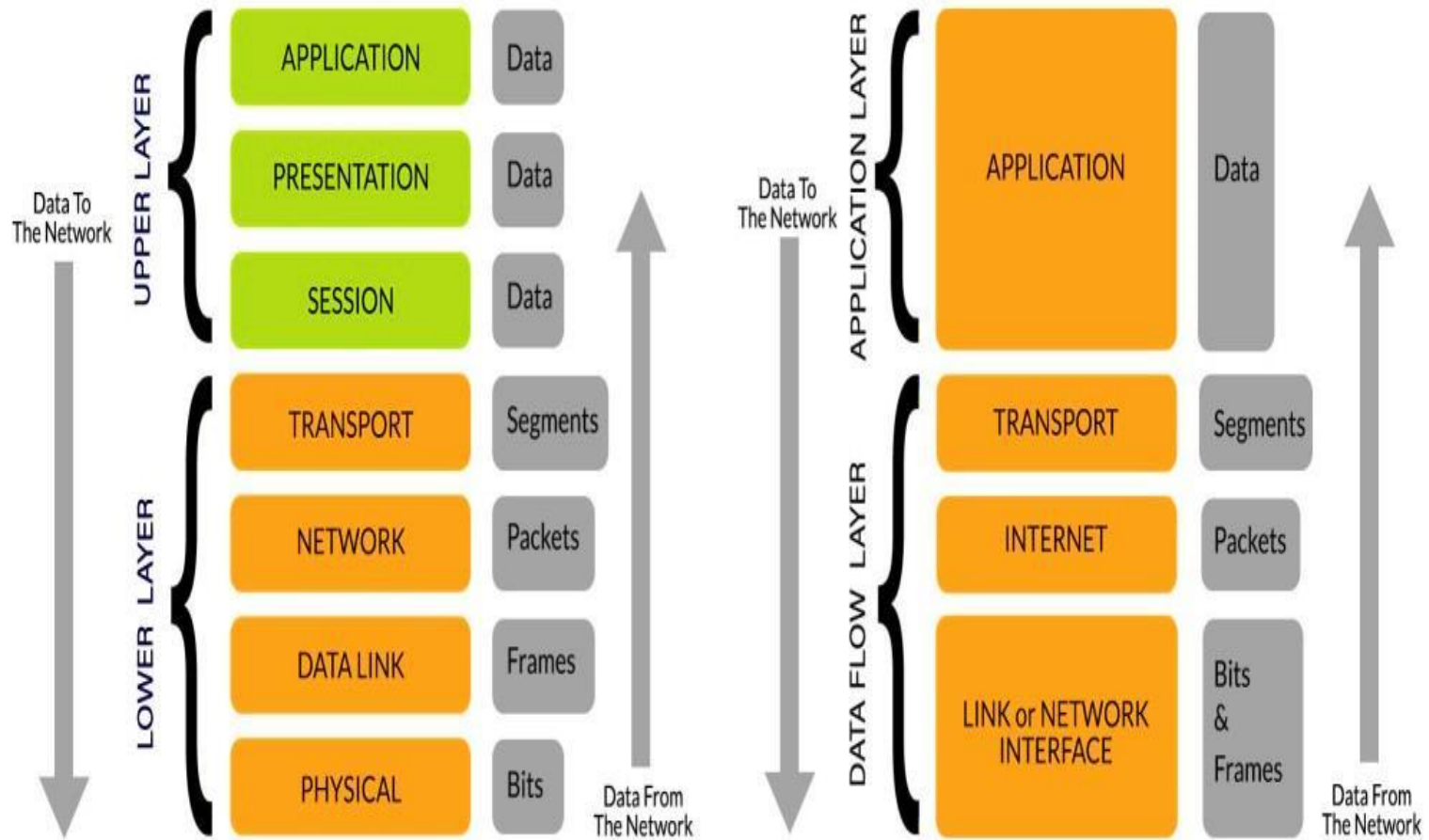
There are **n** numbers of users who use computer network and are located over the world. So to ensure, national and worldwide data communication, systems must be developed which are compatible to communicate with each other ISO has developed a standard. ISO stands for **International organization of Standardization**.

This is called a model for **Open System Interconnection** (OSI) and is commonly known as OSI model.

The ISO-OSI model is seven layer architecture. It defines seven layers or levels in a complete communication system. They are:

- Physical Address
- Data link Layer
- Network Layer
- Transport Layer
- Session Layer
- Presentation Layer
- Application Layer

OSI MODEL vs TCP/IP MODEL



Physical layer

Physical layer is the lowest layer of the OSI reference model. It is responsible for sending bits from one computer to another. This layer is not concerned with the meaning of the bits and deals with the setup of physical connection to the network and with transmission and reception of signals.

Functions of Physical Layer

Following are the various functions performed by the Physical layer of the OSI model.

- **Representation of Bits:** Converts bits into electrical, optical, or radio signals
Example: 0 → low voltage, 1 → high voltage
- **Data Rate:** This layer defines the rate of transmission which is the number of bits per second.
- **Synchronization:** It deals with the synchronization of the transmitter and receiver. The sender and receiver are synchronized at bit level. Sender and receiver clocks must match
- **Transmission medium :** Defines media like: Twisted pair cable, Coaxial cable, Optical fiber, Wireless (radio waves)
- **Line Configuration:** This layer connects devices with the medium: Point to Point configuration and Multipoint configuration.
- **Topologies:** Devices must be connected using the following topologies: Mesh, Star, Ring and Bus.
- **Transmission Modes:** Physical Layer defines the direction of transmission between two devices: Simplex, Half Duplex, Full Duplex.

Data Link Layer

The main task of the **data link layer** is to transform a raw transmission facility into a line that appears free of undetected transmission errors to the network layer. It accomplishes this task by having the sender break up the input data into **data frames**(typically a few hundred or few thousand bytes) and transmit the frames sequentially. If the service is reliable, the receiver confirms correct receipt of each frame by send back an **acknowledgement frame**.

Functions of Data Link Layer

- **Framing:** Frames are the streams of bits received from the network layer into manageable data units. This division of stream of bits is done by Data Link Layer.
- **Physical Addressing:** The Data Link layer adds a header to the frame in order to define physical address of the sender or receiver of the frame, if the frames are to be distributed to different systems on the network.
- **Flow Control:** A flow control mechanism to avoid a fast transmitter from running a slow receiver by buffering the extra bit is provided by flow control. This prevents traffic jam at the receiver side.
- **Error Control:** Error control is achieved by adding a trailer at the end of the frame. Duplication of frame is also prevented by using this mechanism. Data Link Layers adds mechanism to prevent duplication of frames.
- **Access Control:** Protocols of this layer determine which of the devices has control over the link at any given time, when two or more devices are connected to the same link.

Network Layer

The main aim of this layer is to deliver packets from source to destination across multiple links (networks). If two computers (system) are connected on the same link, then there is no need for a network layer.

- It routes the signal through different channels to the other end and acts as a network controller.
- It also divides the outgoing messages into **packets** and to assemble incoming packets into messages for higher levels.

Functions of Network Layer

- It translates logical network address into physical address. Concerned with circuit, message or packet switching.
- Routers and gateways operate in the network layer. Mechanism is provided by Network Layer for routing the packets to final destination.
- Connection services are provided including network layer flow control, network layer error control and packet sequence control.
- Breaks larger packets into small packets.

Transport Layer

The basic function of the Transport layer is to accept data from the layer above, split it up into smaller units, pass these data units to the Network layer, and ensure that all the pieces arrive correctly at the other end.

Functions of Transport Layer

- **Service Point Addressing:** Transport Layer header includes service point address which is port address. This layer gets the message to the correct process on the computer unlike Network Layer, which gets each packet to the correct computer.
- **Segmentation and Reassembling:** A message is divided into segments; each segment contains sequence number, which enables this layer in reassembling the message. Message is reassembled correctly upon arrival at the destination and replaces packets which were lost in transmission.
- **Connection Control:** It includes 2 types:
 - Connectionless Transport Layer: Each segment is considered as an independent packet and delivered to the transport layer at the destination machine.
 - Connection Oriented Transport Layer: Before delivering packets, connection is made with transport layer at the destination machine.
- **Flow Control:** In this layer, flow control is performed end to end.
- **Error Control:** Error Control is performed end to end in this layer to ensure that the complete message arrives at the receiving transport layer without any error. Error Correction is done through retransmission.

Session Layer

- Its main aim is to **establish, maintain and synchronize** the interaction between communicating systems.
- Session layer **manages and synchronize** the conversation between two different applications.
- In Session layer, **streams of data are marked** and are **resynchronized properly**, so that the ends of the messages are not cut prematurely and data loss is avoided.

Functions of Session Layer

- **Dialog Control** : This layer allows two systems to start communication with each other in half-duplex or full-duplex.
- **Token Management**: This layer prevents two parties from attempting the same critical operation at the same time.
 - A special control signal called a **token** is used
 - Only the device holding the token is allowed to:
 - Modify data
 - Perform critical operations
 - Other devices must **wait** until the token is released

Eg: Online railway ticket booking

Only one user gets seat confirmation

Token prevents double booking

- **Synchronization** : This layer allows a process to add checkpoints which are considered as synchronization points into stream of data.

Example: If a system is sending a file of 800 pages, adding checkpoints after every 50 pages is recommended. This ensures that 50 page unit is successfully received and acknowledged. This is beneficial at the time of crash as if a crash happens at page number 110; there is no need to retransmit 1 to 100 pages.

Presentation Layer

- The **Presentation Layer** ensures that data is transmitted in such a way that the **receiver can understand and correctly use the information**. It handles the **representation, format, and meaning of data** exchanged between two communicating systems.
- Different systems may use **different languages (syntax), character sets, and data representations**. To enable communication between such heterogeneous systems, the Presentation Layer defines data structures in an **abstract and standardized form**, ensuring compatibility and interoperability.

Need for Presentation Layer

- Communicating systems may use different:
 - Character encoding schemes
 - Data formats
 - Internal data representations
- Without this layer, data may become **misinterpreted or unusable**
- It ensures **platform-independent communication**

Functions of Presentation Layer

Translation

- Before transmission, information in the form of **characters and numbers** must be converted into **bit streams**.

The Presentation Layer:

- Translates data between different encoding methods
- Converts application format to network format and vice versa
- This is necessary because different computers use different encoding methods.
- **Example:**
ASCII ↔ Unicode conversion

Encryption and Decryption

- The Presentation Layer provides **security** by:
- Encrypting data at the **sender side**
- Decrypting data at the **receiver side**
- **Encryption** converts plain text into **cipher text** to prevent unauthorized access.
Decryption converts cipher text back into plain text.
- **Example:**
Passwords and sensitive data used in online banking and secure websites.

Compression

- The Presentation Layer performs **data compression** to reduce the **bandwidth required for transmission**.
- **Primary purpose of compression:**
- Reduce the number of bits to be transmitted
- Improve transmission speed
- Reduce network congestion
- Compression is especially important for transmitting **multimedia data** such as:
Audio, Video, Images, Text files
- **Examples:**
MP3, MP4, ZIP formats

Application Layer

It is the top most layer of OSI Model. Manipulation of data(information) in various ways is done in this layer which enables user or software to get access to the network. Some services provided by this layer includes: **E-Mail, transferring files, distributing the results to user, directory services, network resources, etc.**

The Application Layer contains a variety of protocols that are commonly needed by users. One widely-used application protocol is **HTTP (Hypertext Transfer Protocol)**, which is the basis for the World Wide Web. When a browser wants a web page, it sends the name of the page it wants to the server using HTTP. The server then sends the page back.

Functions of Application Layer

- **Mail Services:** This layer provides the basis for E-mail forwarding and storage.
- **Network Virtual Terminal:** It allows a user to log on to a remote host. The application creates software emulation of a terminal at the remote host. User's computer talks to the software terminal which in turn talks to the host and vice versa. Then the remote host believes it is communicating with one of its own terminals and allows user to log on.
- **Directory Services:** This layer provides access for global information about various services.
- **File Transfer, Access and Management (FTAM):** It is a standard mechanism to access files and manages it. Users can access files in a remote computer and manage it. They can also retrieve files from a remote computer.

QUALITY OF SERVICE (QoS)

- **Quality of Service (QoS)** is a set of techniques used in computer networks to **control and manage network traffic** so that important applications get better performance than less important ones.
- In networks where many users share the same resources, QoS ensures **reliable, predictable, and efficient data delivery**.

Why QoS is Important

- Modern networks carry different types of data:
 - Voice calls
 - Video conferencing
 - Online streaming
 - Normal data (emails, file downloads)
- All traffic cannot be treated the same. QoS **prioritizes time-sensitive traffic** like voice and video to avoid delay and loss.

Quality of Service Parameters:

- **Bandwidth**

Bandwidth is the **maximum data rate** a network can support.

Measured in **bps, Mbps, or Gbps**.

Higher bandwidth means more data can be transmitted at a time.

Example: Video streaming needs **high bandwidth**.

- **Delay (Latency)**

Delay is the **time taken for data to travel** from source to destination.

Measured in **milliseconds (ms)**.

Low delay is critical for **voice calls and video conferencing**.

- **Jitter**

Jitter is the **variation in delay** between consecutive packets.

Even if delay is low, high jitter causes **uneven packet arrival**.

Affects **VoIP and live video** badly.

Example: Voice breaks or sounds robotic due to jitter.

- **Packet Loss**

Packet loss occurs when **data packets are dropped** during transmission.

Happens due to congestion, buffer overflow, or errors.

Measured as a **percentage (%)**.

High packet loss causes:

- Distorted audio

- Video freezing

Techniques to Improve QoS

Some techniques that can be used to improve the quality of service. The four common methods:

- Scheduling
- Traffic shaping
- Admission control
- Resource reservation.

a. Scheduling

- **Scheduling** is a QoS mechanism that decides **which packet should be transmitted next** when multiple packets are waiting in a queue.
- It helps control **delay, jitter, and fairness** in the network.
- Types:
 - FIFO queuing
 - Priority queuing
 - Weighted fair queuing.

FIFO Queuing

- **FIFO Queuing** is a packet scheduling technique in which **packets are transmitted in the exact order in which they arrive** at the queue.
- The **first packet that enters the queue is the first one to leave**.

How FIFO Queuing Works

- Packets arrive at the router or switch.
- All packets are placed in **a single queue**.
- No priority or classification is applied.
- Packets are transmitted **sequentially**, based on arrival time.

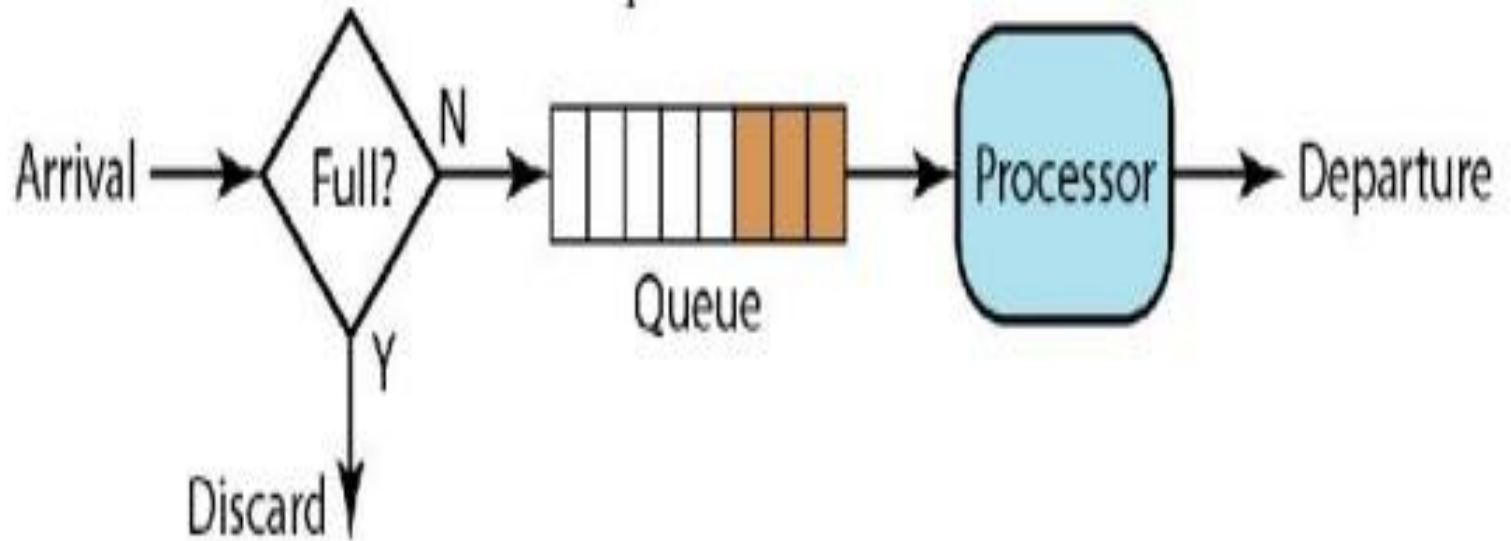
Think of it like a **line at a ticket counter**

Characteristics of FIFO Queuing

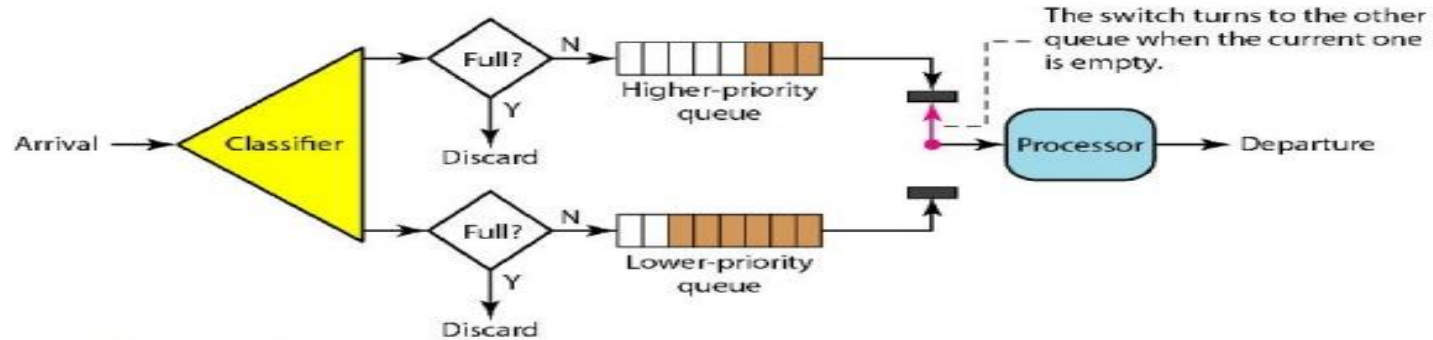
- Uses **only one queue**
- **No priority** for any traffic
- Simple and low-cost
- All packets are treated **equally**

Advantages

- ☐ Easy to implement
- ☐ Low processing overhead
- ☐ Suitable for small or lightly loaded networks



Priority Queuing



Priority Queuing is a scheduling technique in which **packets are classified into different priority levels**, and **higher-priority packets are transmitted before lower-priority packets**.

Common priority levels: **High, Medium, Low, Very Low**

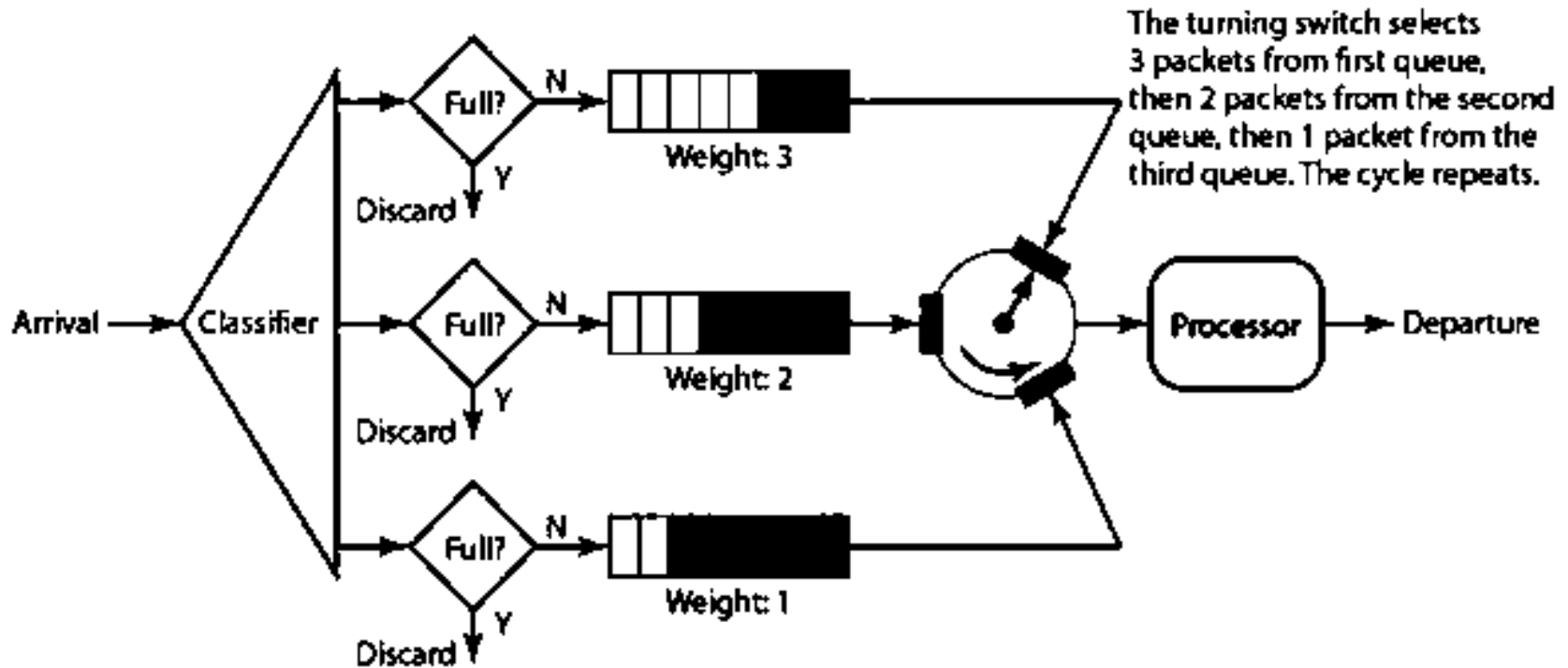
How Priority Queuing Works

- Incoming packets are **classified** based on type (voice, video, data).
- Packets are placed into **separate queues** according to priority.
- The scheduler always checks the **highest-priority queue first**.
- Lower-priority queues are served **only when higher ones are empty**.

Example: **Ambulance gets priority over normal traffic**

- However, there is a potential drawback. If there is a continuous flow in a high-priority queue, the packets in the lower-priority queues will never have a chance to be processed. This is a condition called **starvation**

Weighted Fair Queuing



Weighted Fair Queuing (WFQ) is a packet scheduling algorithm in which **each queue is assigned a weight**, and bandwidth is distributed **proportionally based on that weight**.

Higher weight → More bandwidth

Lower weight → Less bandwidth

Why WFQ is Needed?

- FIFO → No priority
- Priority Queuing → Starvation problem
- WFQ → Provides **both fairness and priority**

How WFQ Works

- Traffic is classified into different flows or queues.
- Each queue is assigned a **weight value**.
- Scheduler distributes bandwidth based on weight.
- All queues get service — no starvation.

Example:

If three queues have weights 3:2:1

b. Traffic Shaping

Traffic shaping is a mechanism to control the amount and the rate of the traffic sent to the network.

Two techniques can shape traffic:

- Leaky bucket
- Token bucket

Leaky bucket Algorithm

Imagine a bucket with:

- A **small hole at the bottom**
- Water poured into it at **any speed**
- Water leaking out at a **constant speed**

Even if water is poured fast (burst),

The output flow is always smooth and constant.

- If too much water is poured and the bucket becomes full →
Extra water **overflows (lost)**.

Why Leaky Bucket is Needed?

In real networks:

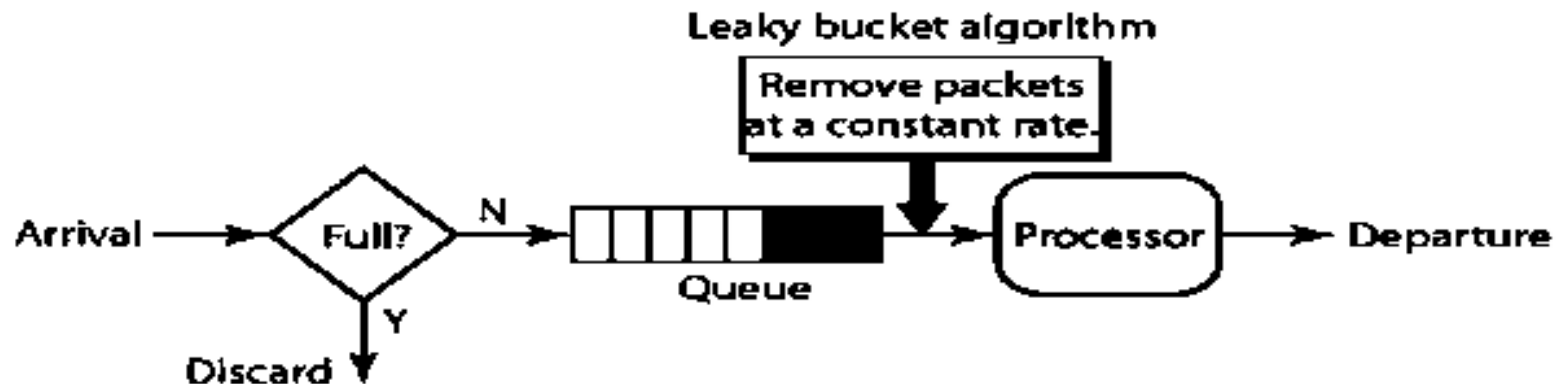
- Users send data in **bursts**
- Bursty traffic causes:
 - Congestion
 - Packet loss
 - High delay
 - Jitter

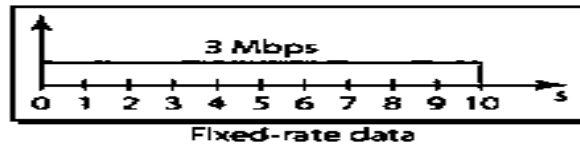
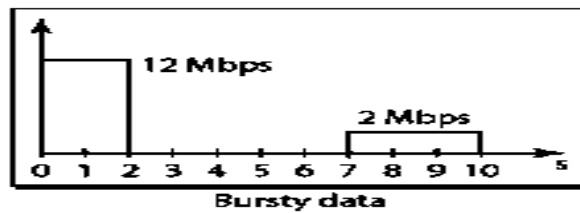
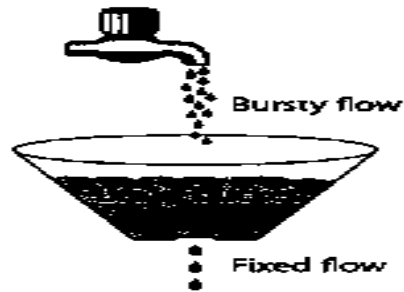
So we use Leaky Bucket to:

- Convert **bursty input** → **smooth output**
- Enforce committed bandwidth
- Protect the network

The Leaky Bucket consists of:

- **Bucket = Buffer (Queue)**
- **Hole = Fixed output rate**
- **Water = Packets**
- **Overflow = Packet drop**
- **Step-by-Step Working:**
 - Packets arrive at variable rate.
 - They are stored in a **FIFO queue**.
 - At every clock tick, packets are removed at a **fixed rate**.
 - If queue becomes full:
 - New arriving packets are **discarded**.





Network committed bandwidth = **3 Mbps**

Step 1: Burst Transmission

Host sends **12 Mbps** for **2 seconds**

Total data = $12 \times 2 = 24$ **Mbits**

Step 2: Silent Period

Host silent for **5 seconds**

Step 3: Sends Again

2 Mbps for **3 seconds**

Total = $2 \times 3 = 6$ **Mbits**

Total Data Sent

$24 + 6 = 30$ **Mbits in 10 seconds**

Output of Leaky Bucket

Bucket leaks at constant **3 Mbps**

So in 10 seconds:

$3 \times 10 = 30$ **Mbits sent**

Input was bursty

Output becomes smooth at 3 Mbps

Token Bucket Algorithm

The **Token Bucket Algorithm** is a traffic shaping and traffic policing mechanism used in computer networks to control the amount of data transmitted into the network.

Unlike Leaky Bucket, Token Bucket:

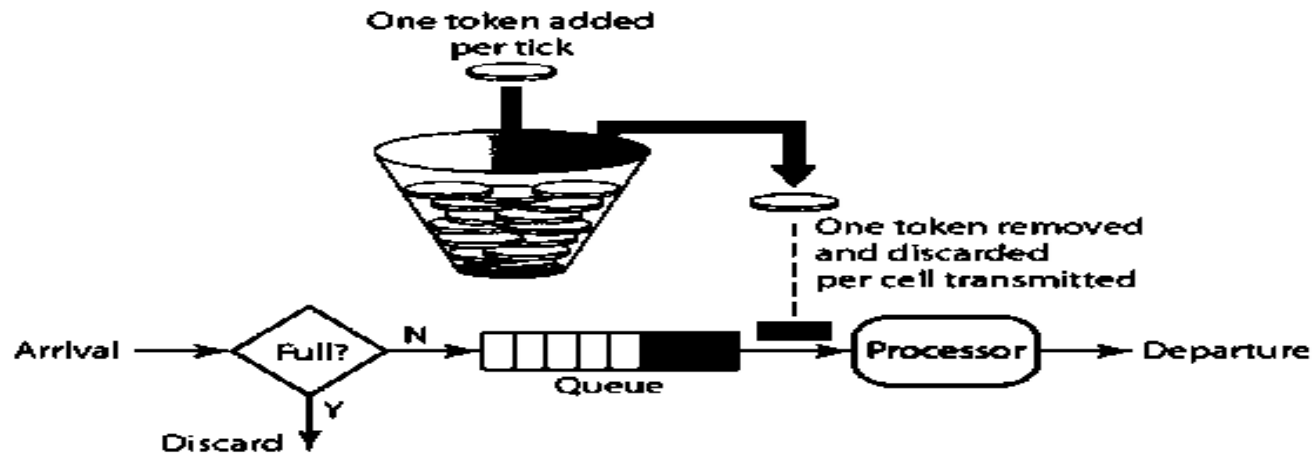
- Allows burst transmission
- Accumulates credit when idle
- Provides flexibility

It is widely used in QoS mechanisms.

Basic Concept

Think of a bucket that stores **tokens** instead of packets.

- Tokens are added to the bucket at a fixed rate.
- To send data, the host must remove tokens.
- If no tokens are available → data cannot be sent.
- If bucket is full → extra tokens are discarded (not packets).
- Tokens represent permission to send data.



Token Generation (Top Part of Diagram)

- At the top, you see:
 - ☐ "One token added per tick"
- Tokens are generated at a **fixed rate** (for example, 1 token per clock tick).
- These tokens are stored in the **token bucket**.
- If the bucket becomes full, extra tokens are discarded.
- Tokens represent permission to send data.

Arrival of Data (Left Side)

- On the left side:
- Arrival → "Full?" decision box
- When data packets arrive:
 - If queue is **not full (N)** → packets enter the queue.
 - If queue is **full (Y)** → packets are discarded.
- So this part controls **buffer overflow**.

Queue

- The middle rectangular boxes represent the **Queue (FIFO)**.
- Incoming packets wait here.
- They cannot be transmitted unless tokens are available.

Token Removal During Transmission

- Near the processor, the diagram shows:
- ☐ “One token removed and discarded per cell transmitted”
- This means:
- For each packet (or byte) transmitted,
- One token is removed from the bucket.
- If tokens are available → packet is sent.
- If no tokens → transmission stops.

Processor and Departure

- After token verification:
- Queue → Processor → Departure
- Processor sends packets out of the system.
- Output depends on available tokens.

Difference from Leaky Bucket

- Leaky Bucket:
 - ☐ Does not allow burst
 - ☐ Idle time not considered
- Token Bucket:
 - ☐ Allows burst
 - ☐ Idle time accumulates tokens
 - ☐ More flexible

c. Resource Reservation

A data flow (like video call, live streaming) needs network resources such as:

- **Buffer space** (to store packets temporarily)
- **Bandwidth** (data transmission capacity)
- **CPU time** (processing power in routers)
- **Queue space**

If these resources are reserved in advance, the network can guarantee better Quality of Service (QoS).

d . Admission Control

Admission Control is a mechanism used by a router or switch to **accept or reject a new data flow** based on available network resources.

Before allowing a new flow, the router checks whether it has enough:

- Bandwidth
- Buffer space
- CPU processing power
- Queue capacity

If sufficient resources are available → flow is accepted.

If not → flow is rejected.

APPLICATION LAYER PROTOCOLS

The application layer in the OSI model is the closest layer to the end user which means that the application layer and end user can interact directly with the software application. The application layer programs are based on client and servers.

DNS

- DNS stands for **Domain Name System**.
- DNS is a **directory service** that provides a mapping between the **name of a host** on the network and its **numerical address**.
- DNS is required for the functioning of the internet.
- Each **node in a tree** has a domain name, and a full domain name is a sequence of symbols specified by dots.
- DNS is a service that **translates the domain name into IP addresses**. This allows the users of networks to utilize user-friendly names when looking for other hosts instead of remembering the IP addresses.
- For example, suppose the FTP site at EduSoft had an IP address of 132.147.165.50, most people would reach this site by specifying ftp.EduSoft.com. Therefore, the domain name is more reliable than IP address.
- DNS is a TCP/IP protocol used on different platforms.

The domain name space is divided into three different sections:

- Generic Domains
- Country Domains
- Inverse Domain.

Generic Domains

- It defines the registered hosts according to their **generic behavior**.
- Each node in a tree defines the domain name, which is an index to the DNS database.
- It **uses three-character labels**, and these labels describe the organization type.

Label	Description
aero	Airlines and aerospace companies
biz	Businesses or firms
com	Commercial Organizations
coop	Cooperative business Organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International Organizations
mil	Military groups
museum	Museum & other nonprofit organizations
name	Personal names
net	Network Support centers
org	Nonprofit Organizations
pro	Professional individual Organizations

Country Domain

The format of country domain is same as a generic domain, but it **uses two-character country abbreviations** (e.g., us for the United States) in place of three character organizational abbreviations

Inverse Domain

The inverse domain is used for **mapping an address to a name**. When the server has **received a request** from the client, and the **server contains the files of only authorized clients**. To determine **whether the client is on the authorized list or not**, it **sends a query to the DNS server and ask for mapping** an address to the name.

Working of DNS

- DNS is a client/server network communication protocol.
- DNS clients send requests to the server while DNS servers send responses to the client.
- Client requests **contain a name** which is **converted into an IP address** known as a **forward DNS lookups** while requests containing an IP address which is converted into a name known as **reverse DNS lookups**.
- DNS implements a **distributed database to store the name of all the hosts available on the internet**.
- If a client like a web browser sends a request containing a hostname, then a piece of software such as **DNS resolver** sends a request to the DNS server to obtain the IP address of a hostname. If DNS **server does not contain** the IP address associated with a hostname, then it forwards the request to **another DNS server**. If IP address has arrived at the resolver, which in turn completes the request over the internet protocol.

FTP

- **FTP (File Transfer Protocol)** is a standard internet protocol used to **transfer files between two computers** over a network (internet).

☐ It is mainly used for:

- Uploading web pages to a server.
- Downloading files from servers.
- Sharing files between computers.

Example:

Uploading a website from your computer to a web server.

Objectives of FTP

- **File sharing** between computers.
- **Remote access** — access files from another computer.
- **Reliable and efficient data transfer.**

Why FTP?

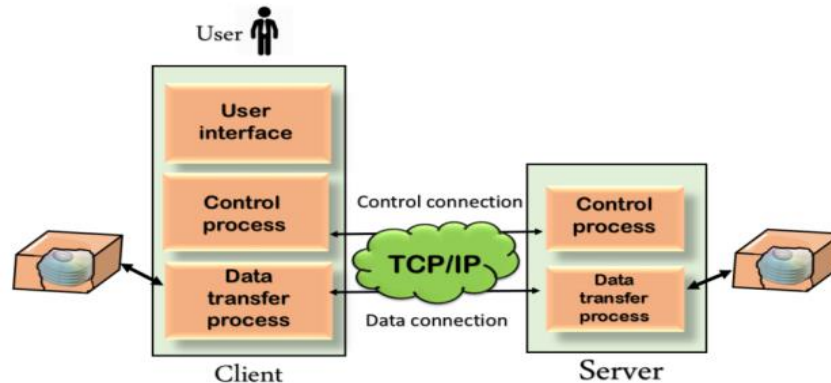
- Different computers may have:
- Different file formats
- Different directory structures
- Different data representation methods

☐ FTP solves these problems by creating two connections:

Control connection → for commands.

Data connection → for file transfer.

Mechanism of FTP



FTP follows a **Client–Server model**.

FTP Client (User Side)

- Has 3 parts:
- **User Interface** → where user gives commands.
- **Control Process** → manages communication.
- **Data Transfer Process** → sends/receives files.

FTP Server

- Has 2 parts:
- Server Control Process
- Server Data Transfer Process
-  Client requests → Server responds → File transferred.

Control Connection

- Used to send commands.
- Example: login, file request.
- Remains active throughout session.
- Simple communication.
- ☐ Like giving instructions.

Data Connection

- Used to transfer actual files.
- Opens when file transfer starts.
- Closes after transfer completes.
- ☐ Like sending the file itself.

Advantages of FTP

- ✓ Very fast file transfer.
- ✓ Efficient — easy data sharing.
- ✓ Requires login (username & password).
- ✓ Supports sending and receiving files repeatedly.

❓ Disadvantages of FTP

- ❓ Data may not be encrypted (security risk).
- ❓ Password sent in plain text → hacking possible.
- ❓ File size limitations.
- ❓ Not compatible with all systems.

HTTP

- **HTTP (HyperText Transfer Protocol)** is a communication protocol used to access data on the **World Wide Web (WWW)**.
 - It transfers data between:
 - **Client** → Browser (Chrome, Firefox)
 - **Server** → Website server
 - It can transfer:
 - Text
 - Images
 - Audio
 - Video
 - Hypertext documents
- ☐ Example: When you open Google, your browser sends an HTTP request and Google sends a response.

Features of HTTP:

Connectionless Protocol

HTTP creates a connection only for a **single request and response**. After the server sends the response, the connection is **closed automatically**.

Working:

- Client sends request.
- Server processes request.
- Server sends response.
- Connection ends.
- ☐ The client and server are connected **only during communication**.

Example:

- You open a webpage → page loads → connection closes.

Advantage:

- Saves network resources and memory.

Media Independent

- HTTP can transfer **any type of data**.
- Client and server must understand the data format.

Types of data HTTP can send:

- Text
- Images
- Audio
- Video
- Files

How it works:

- Uses **MIME type (Multipurpose Internet Mail Extensions)** to specify data type.

Example MIME Types:

- text/html → webpage
- image/jpeg → image
- video/mp4 → video
- ❓ So HTTP is not restricted to one type of content.

Stateless Protocol

- HTTP does **not remember previous requests**.
- Each request is treated as a **new request**.

Working:

- Client and server know each other only during current communication.
- No information is stored between requests.

Example:

- After login, websites use **cookies** to remember you because HTTP itself cannot remember.

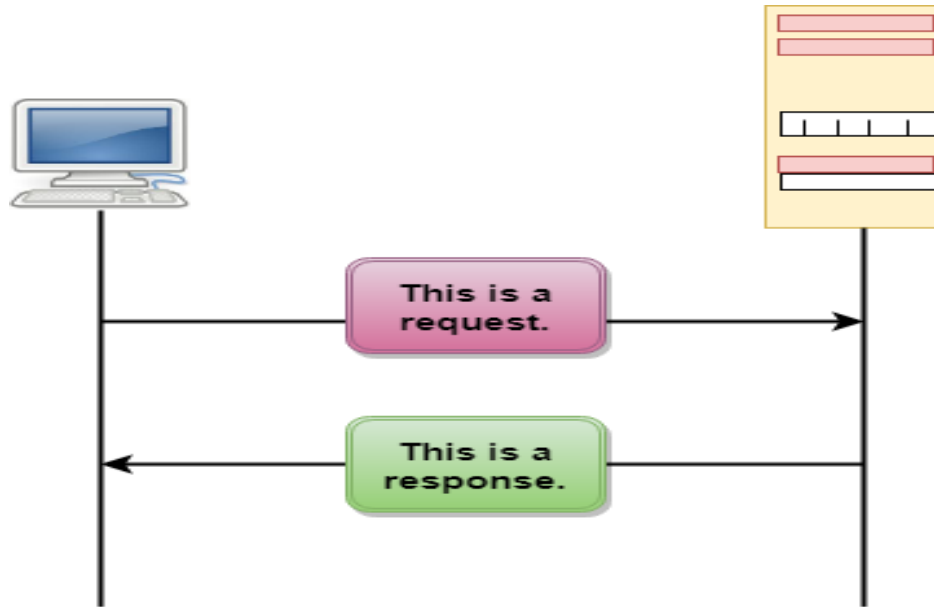
Advantage:

- Simple and fast communication.
- Less server memory needed.

Limitation:

- Cannot maintain user session by itself.

HTTP Transactions



Client sends request to server.

Server processes request.

Server sends response.

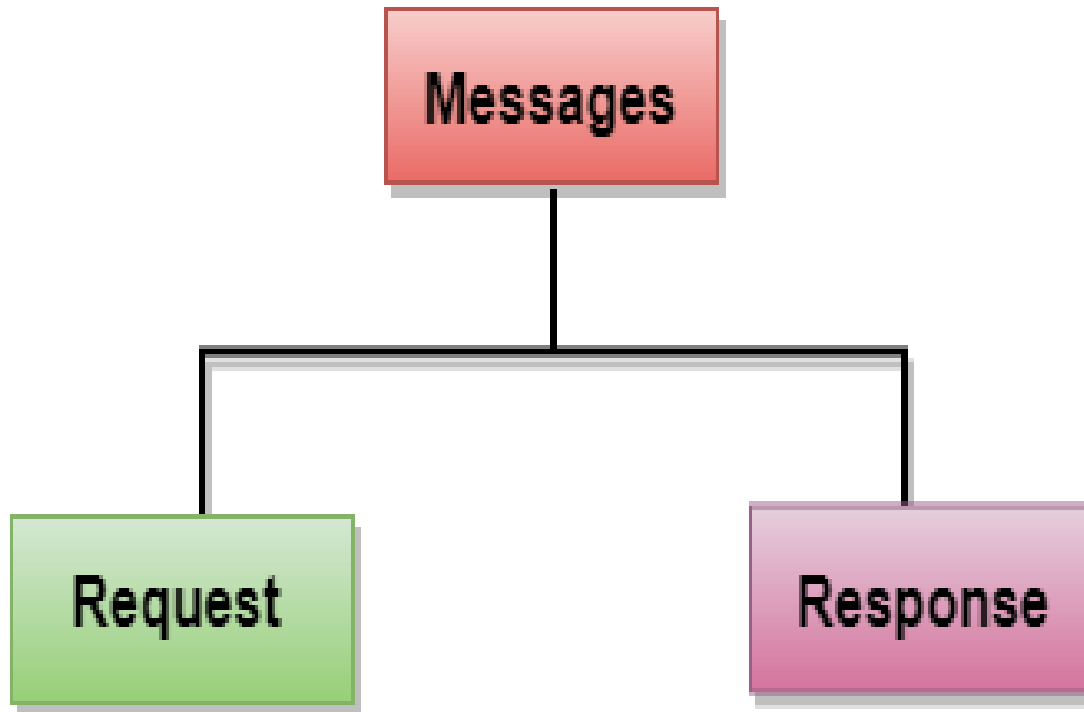
Example:

You search YouTube → request

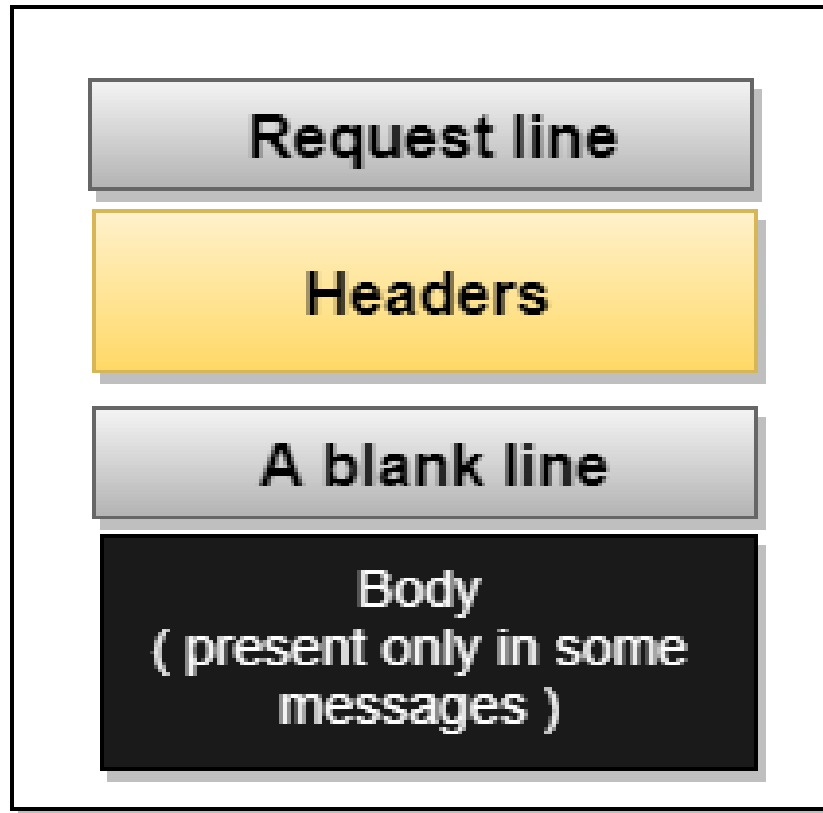
YouTube page loads → response

Messages

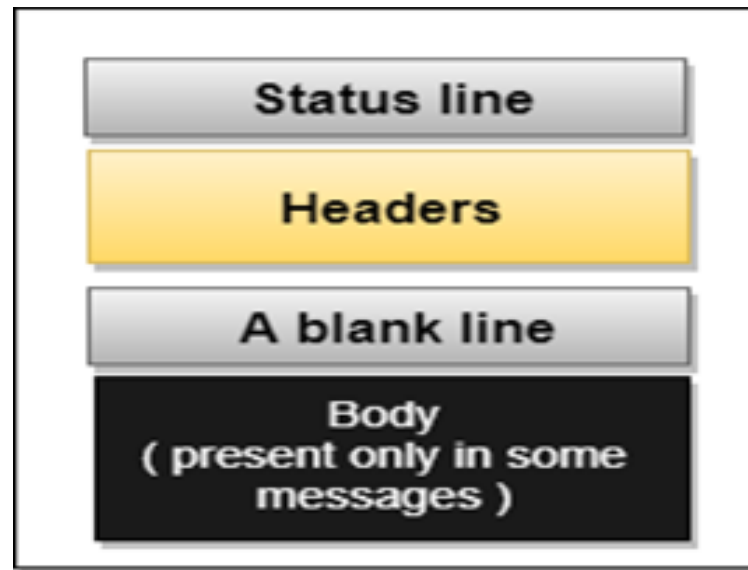
- HTTP messages are of two types: request and response. Both the message types follow the same message format.



- **Request Message:** The request message is sent by the client that consists of a request line, headers, and sometimes a body.



- **Response Message:** The response message is sent by the server to the client that consists of a status line, headers, and sometimes a body.



Uniform Resource Locator (URL)

- Address used to locate a resource on internet.
- Standard way to access web documents.
- The URL defines four parts: method, host computer, port, and path.



A URL has 4 parts:

1. Method / Protocol

- How data is retrieved.
- Example: HTTP, HTTPS, FTP.

2. Host

- Server name or domain.
- Example: www.google.com

3. Port (Optional)

- Communication channel.
- Default HTTP port = 80.

4. Path

- Location of file on server.

☐ Example:

- `http://www.example.com:80/index.html`

Round Trip Time (RTT)

Round Trip Time (RTT) is the total time taken by a data packet to:

- travel from **client** → **server**, and
- return from **server** → **client**.
- ☐ It measures **network delay (latency)**.

How RTT Works

- Client sends a request to server.
- Server receives and processes request.
- Server sends response back.
- Total time taken = **RTT**.

Example

- You click a website link.
- Request goes to server.
- Server sends webpage.
- Time taken for full cycle = RTT.

TCP 3-Way Handshake is the process of establishing a reliable connection between client and server before data transfer.

- It ensures both systems are ready to communicate.

Steps of TCP 3-Way Handshake

Step 1: Connection Request (SYN)

- Client asks server to establish connection.
- Sends **SYN (synchronize)** message.
- ☐ “Can we connect?”

Step 2: Server Response (SYN-ACK)

- Server accepts request.
- Sends **SYN + ACK (acknowledgment)**.
- ☐ “Yes, connection allowed.”

Step 3: Client Acknowledgment (ACK)

- Client confirms server response.
- Sends ACK.
- ☐ “Connection confirmed.”
- Now connection is established and data transfer begins.

Real-Life Example

- Like making a phone call:
- You call someone → request.
- They answer → response.
- You say hello → confirmation.
- Now conversation starts.

Types of HTTP Connections

HTTP connections are categorized into two types

- ***Non-Persistent*** –
 - Without a Parallel Connection
 - With Parallel Connection
- ***Persistent-***
 - Non-Pipelined
 - Pipelined

Non-Persistent

- In **non-persistent HTTP**, a **new TCP connection is created every time** a client (browser) requests an object (like an image, webpage, or file).

How it works

- Browser requests a webpage.
- **A TCP connection opens.**
- Server sends the data.
- Connection **closes**.
- If another object is needed → **new connection opens again.**
- ☐ So for one webpage with many items (images, CSS, videos), many connections are opened and closed.

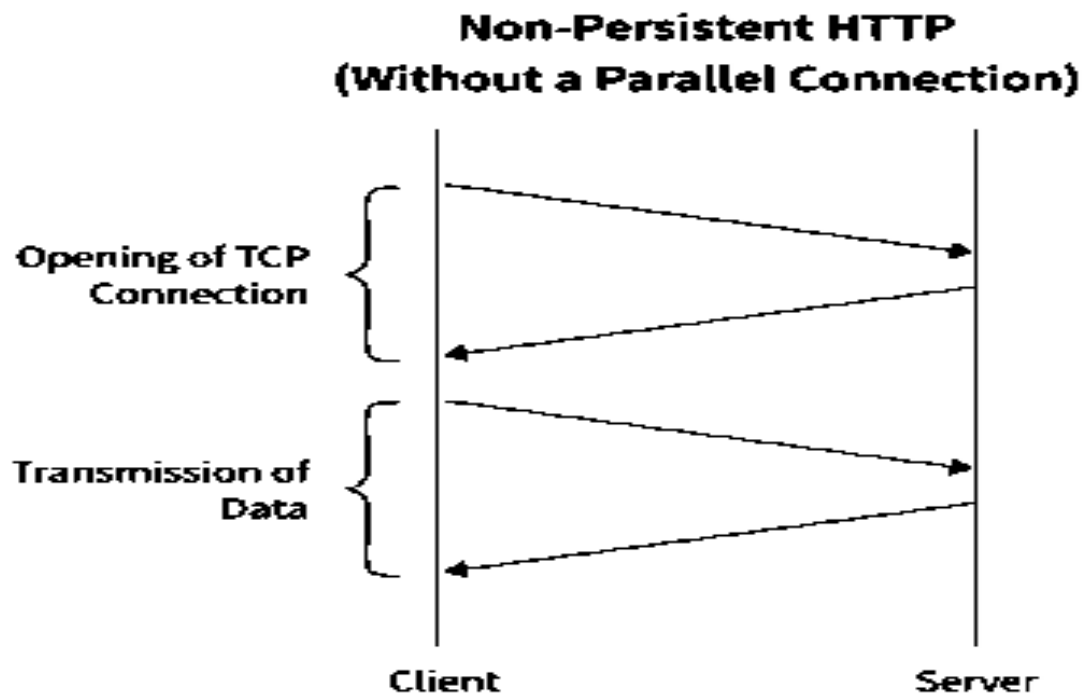
Problem (Overhead)

Opening a TCP connection each time causes:

- Extra time delay
- More network traffic
- Slower communication
- This extra work is called **overhead**.

Without a Parallel Connection

In an HTTP connection without a parallel connection, each transmission takes 2 RTT (Round Trip Time) to be transmitted. The first RTT is to open a TCP connection and the second RTT corresponds to the transmission of the object on the connection



With Parallel Connection

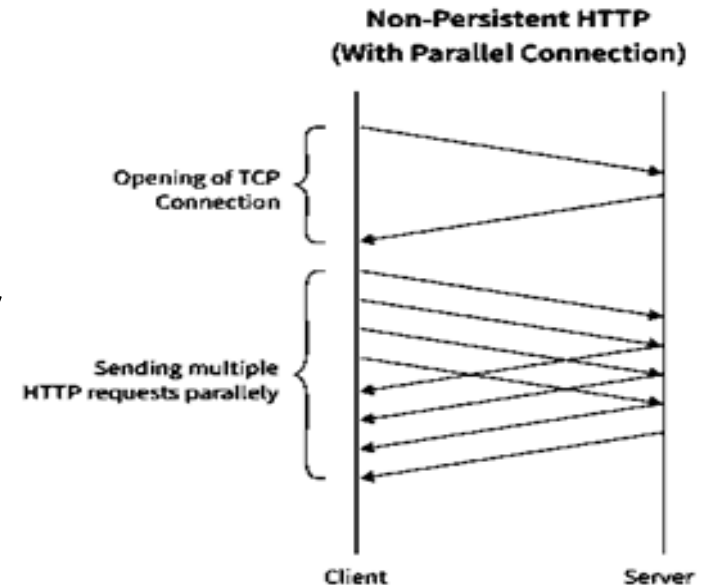
To reduce delay, browsers sometimes use **parallel connections**.

What happens here

- Browser opens **multiple TCP connections at the same time**.
- Many HTTP requests are sent simultaneously.
- Multiple objects download together.

Result

- ? Saves time
- ? Faster webpage loading
- ? Reduces repeated connection setup delay



Persistent

In **persistent HTTP**, a **single TCP connection is kept open** and used to send multiple HTTP requests and responses.

This avoids opening a new connection each time (unlike non-persistent HTTP).

Persistent HTTP has **two types**:

- **Non-Pipelined Persistent HTTP**
- **Pipelined Persistent HTTP**

Persistent HTTP — Non-Pipelined

- One TCP connection is used.
- But requests are sent **one after another**.
- The client must **wait for one response before sending the next request**.

How it works

- TCP connection opens once.
- Client sends request for object 1.
- Server sends response.
- Client sends request for object 2.
- Server sends response.
- Continues like this.
- ☐ **Sequential process (wait → request → wait → request).**

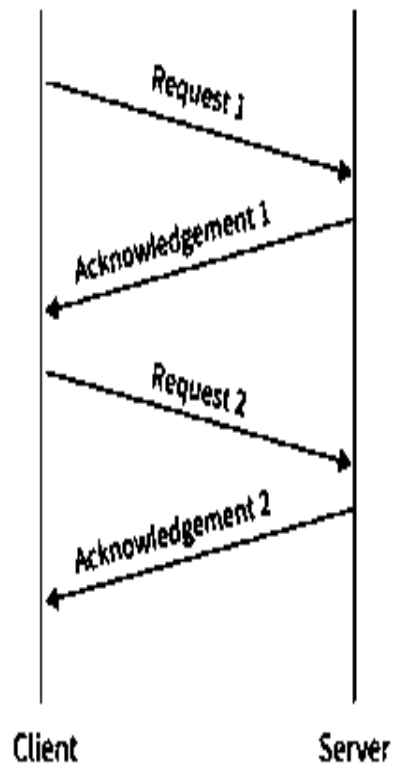
Persistent HTTP — Pipelined

- One TCP connection is used.
- Client sends **multiple requests at once** without waiting.
- Server sends responses in order.
- [?] Requests are sent continuously like a pipeline.

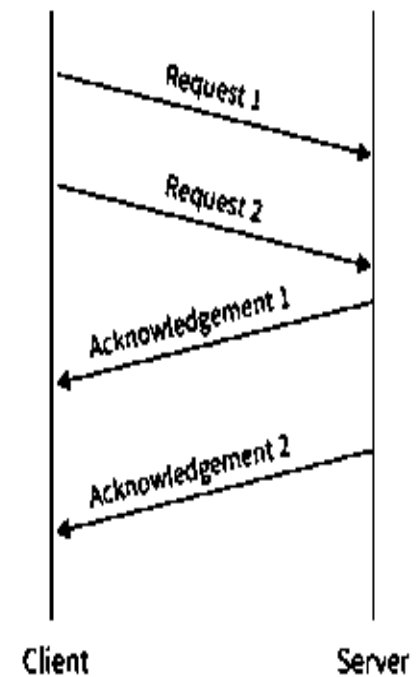
How it works

- TCP connection opens once.
- Client sends request for object 1, 2, 3 together.
- Server sends responses one by one.
- [?] **No waiting between requests → faster.**

Persistent HTTP (Non-Pipelined)



Persistent HTTP (Pipelined)



POP PROTOCOL

- **POP** stands for **Post Office Protocol**.
- The latest version is **POP3 (Post Office Protocol version 3)**.
- It is used to **receive emails from a mail server to a user's computer**.
- It is part of the **Message Access Agent**.
- **POP downloads emails from the server to your device**.

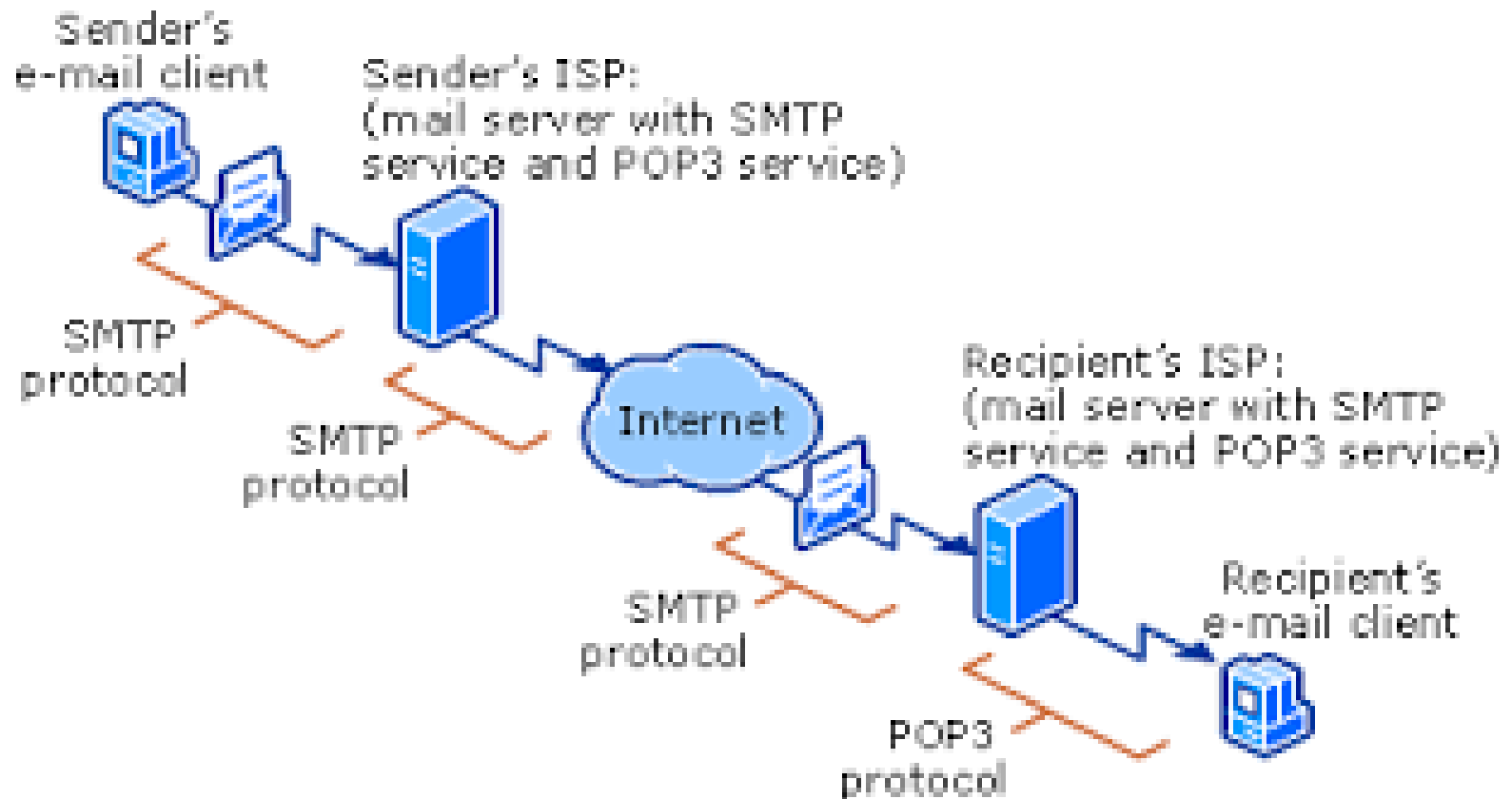
Email communication uses two types of protocols:

- **SMTP (Simple Mail Transfer Protocol)** → sends email (push protocol).
 - **POP3** → receives email (pull protocol).
-
- ✓ **SMTP** → Client → Sender Mail Server → Receiver Mail Server
 - ✓ **POP3** → Receiver Mail Server → Receiver (user device)

So:

- **SMTP = Sending mail**
- **POP3 = Receiving mail**

How Mail is Transmitted

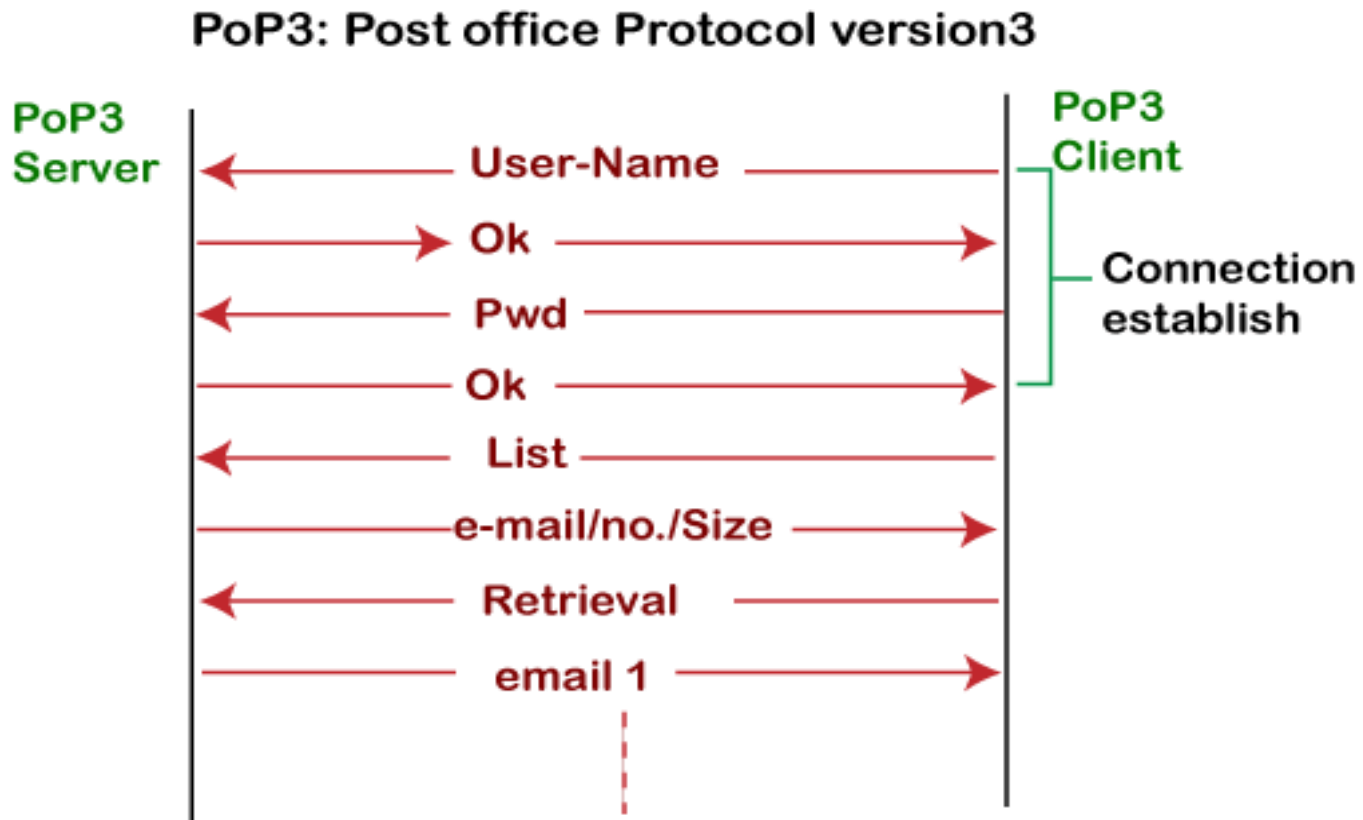


- **Sender writes email** in mail client.
- Email goes to **sender's mail server** using SMTP.
- Sender's server sends mail to **receiver's mail server** through internet.
- Email is stored in receiver's mail server.
- Receiver's device downloads mail using **POP3 or IMAP**.

Characteristics of POP3:

- Simple protocol with limited features.
- Emails are downloaded to local system.
- After download, emails are usually deleted from server.
- Read emails **offline** after downloading.

Let's understand the working of the POP3 protocol.



Step-by-step working:

- **Connection established** between client and POP3 server.
- Server asks for **username**.
- Client sends username → server verifies.
- Server asks for **password**.
- Client sends password → verified.
- Server sends list of emails (number and size).
- Client selects and downloads emails.
- Emails are deleted from server (by default).
- Emails become available only on that computer unless copy is kept on server.

- To establish the connection between the POP3 server and the POP3 client, the POP3 server asks for the user name to the POP3 client.
- If the username is found in the POP3 server, then it sends the ok message.
- It then asks for the password from the POP3 client; then the POP3 client sends the password to the POP3 server.
- If the password is matched, then the POP3 server sends the OK message, and the connection gets established.
- After the establishment of a connection, the client can see the list of mails on the POP3 mail server. In the list of mails, the user will get the email numbers and sizes from the server. Out of this list, the user can start the retrieval of mail.
- Once the client retrieves all the emails from the server, all the emails from the server are deleted.
- Therefore, we can say that the emails are restricted to a particular machine, so it would not be possible to access the same mails on another machine.
- This situation can be overcome by configuring the email settings to leave a copy of mail on the mail server.

Advantages of POP3

- Fast and simple to use.
- Emails available offline.
- Easy configuration.
- No mailbox size limit on server.
- Most emails stored on local computer.

Disadvantages of POP3

- Emails not easily accessible from multiple devices.
- Emails stored locally → risk of virus or data loss.
- Difficult to transfer mail to another system.
- Security risk if system is compromised.
- Needs more local storage space.